Direct and indirect beams of interstellar helium as a calibration-free probe of the ionization rate

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Abstract

Pickup ions measured by New Horizons/SWAP suggest that the ionization rate of interstellar neutral (ISN) helium is about 43% higher than solar-proxy models predict. Pickup-ion and direct-beam measurements cannot tell a higher ionization rate apart from a higher interstellar density, because both scale with their product, and the direct-beam flux also depends on a poorly known detector efficiency. ISN helium reaches 1 au on two kinds of orbits: a *direct* beam and an *indirect* beam that first passes around the Sun. Indirect atoms spend more time close to the Sun and are ionized more, so the ratio of the two beam fluxes depends on the ionization rate but not on the density or on the absolute efficiency. I build a simple analytic model of ISN helium at 1 au and use it to predict both beams over the year as seen by a Sun-pointing spinning detector such as IMAP-Lo. A 43% higher ionization rate lowers the ratio of the late-October indirect beam to the February direct beam by a factor of 0.61, while a 43% higher density leaves it unchanged. Measuring both beams therefore separates the two explanations. I discuss the physical and instrumental effects that a full analysis would need to include.

1 Introduction

The Sun moves through the partially ionized local interstellar medium, and the neutral part of that medium flows freely into the heliosphere. Helium is the best-studied interstellar neutral (ISN) species. It is barely filtered at the heliospheric boundary, and radiation pressure on it is weak, so its atoms move on nearly Keplerian orbits and keep the imprint of the interstellar conditions. The flow was measured by the GAS instrument on Ulysses (Witte, 2004) and, since 2009, by IBEX-Lo on the Interstellar Boundary Explorer (McComas et al., 2009; Fuselier et al., 2009; Möbius et al., 2009). Analyses of the IBEX-Lo data (Bzowski et al., 2012; Lee et al., 2012) led to a consensus set of inflow parameters (McComas et al., 2015), and revealed a slower, warmer secondary population, the "Warm Breeze" (Kubiak et al., 2014).

On the way in, helium atoms are lost mainly through ionization by solar EUV radiation (Sokół et al., 2019). The ionization rate therefore sets how much helium survives to 1 au, and every interpretation of ISN He data depends on it. Two recent results suggest the rate is higher than models based on solar proxies predict. Swaczyna et al. (2022) found that the evolution of the ISN He flux seen by IBEX-Lo over a solar cycle requires a change in detection efficiency, a higher ionization rate, or an additional signal. Antonik et al. (2025) derived the ionization rate from He^+ pickup ions measured by New Horizons/SWAP and found it to be about 43% higher than the model rate.

Neither result can separate the ionization rate from the interstellar density. Pickup-ion production is proportional to the product of density and ionization rate. The direct-beam flux at IBEX-Lo is proportional to the density, to the survival probability and to the absolute detection efficiency, which is only known to tens of percent. "More ionization", "more helium" and "a different efficiency" are therefore largely degenerate.

Bzowski et al. (2023) proposed a way around this: compare the fluxes of the *direct* and *indirect* beams. Indirect atoms pass around the Sun before reaching the detector, so they are exposed to ionization for longer. In the ratio of the two beams the density and the absolute efficiency cancel, while the ionization rate does not. Earlier attempts were limited because the indirect beam could not be separated from the background. IMAP-Lo on the Interstellar Mapping and Acceleration Probe (McComas et al., 2018) sits on a pivot platform that can point at the indirect beam, and Kubiak et al. (2023) designed an observation plan that includes calibration-free determination of the He ionization rate.

This work is a small, independent implementation of the direct/indirect-beam idea. It asks a narrow question: if the SWAP result is right, how large is the change in the beam ratio, and does the ratio really distinguish a higher ionization rate from a higher density?

2 Method

2.1 Model of ISN helium at 1 au

Far from the Sun the helium is a Maxwellian gas moving towards the Sun with the inflow parameters in Table 1. Near the Sun, every atom follows a Keplerian hyperbola under solar gravity; radiation pressure is neglected. Two such orbits reach every point near the Sun: the *direct* orbit, which has swept less than 180° around the Sun since it left infinity, and the *indirect* orbit, which has swept more than 180° (Figure 1).

Along the way atoms are ionized at a rate that falls off with distance from the Sun as

$$\beta(r) = \beta_0 \left(\frac{1\,\mathrm{au}}{r} \right)^2 , \qquad (1)$$

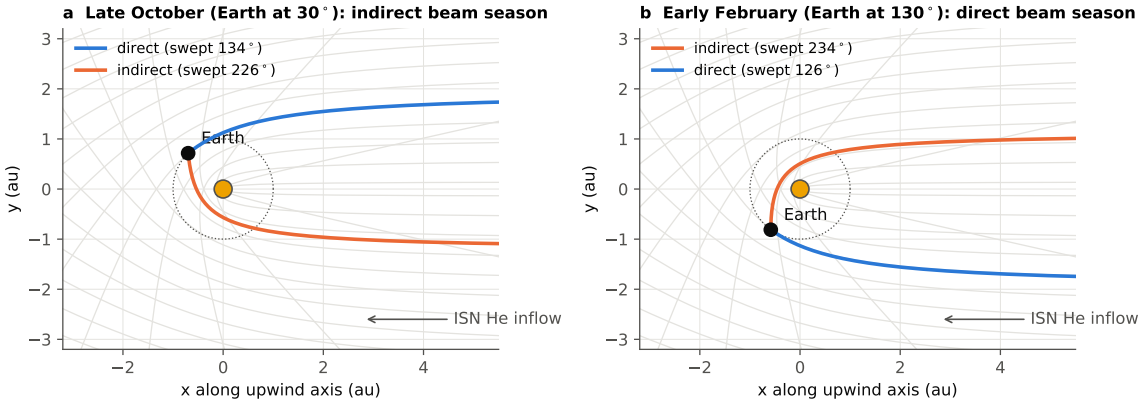


Figure 1: Direct (blue) and indirect (orange) orbits of a cold helium flow reaching Earth in late October (a) and early February (b), in the plane of the inflow. The gas arrives from the right, and the dotted circle is Earth's orbit. The angle each orbit has swept around the Sun is given in the legend.

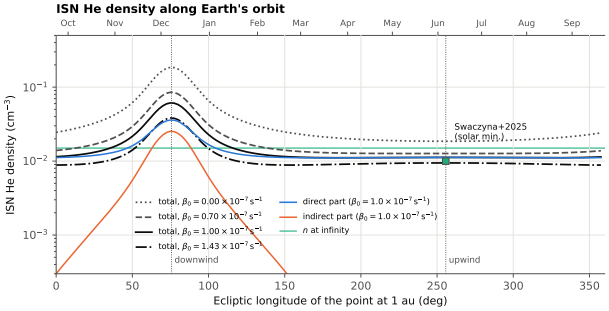


Figure 2: ISN He density along Earth's orbit for several ionization rates, and its direct and indirect parts for $\beta_0 = 1 \times 10^{-7}\,\mathrm{s}^{-1}$. The green line is the density far from the Sun and the square is the solar-minimum value of Swaczyna et al. (2025).

Table 1: Model parameters and reference values. Inflow parameters from McComas et al. (2015); enhancement factor from Antonik et al. (2025).

ISN He far from the Sun	
Density n_{ISN}	$0.015\,\mathrm{cm}^{-3}$
Inflow speed	$25.4\,\mathrm{km\,s}^{-1}$
Upwind longitude, latitude	255.7°, 5.1°
Temperature	7500 K
Ionization	
Model rate at 1 au, β_0	$1.0 \times 10^{-7}\,\mathrm{s}^{-1}$
Enhancement factor (SWAP)	1.43
Observations	
Observer orbit	circular, 1 au
Direct beam	Earth at 130°, early Feb.
Indirect beam	Earth at 30°, late Oct.

where β_0 is the rate at 1 au. For this form of $\beta(r)$ the total exposure along a Keplerian orbit has a simple closed form, so the fraction of atoms that survive to the observer is

$$S = e^{-\beta_0 X}, \qquad X = \frac{(1\,\mathrm{au})^2\,\Delta\nu}{h}, \tag{2}$$

where $\Delta\nu$ is the angle the atom has swept since infinity and h is its angular momentum. The exposure factor X has units of time and depends only on the orbit. Indirect atoms have a larger $\Delta\nu$ and pass closer to the Sun, so their X is several times larger.

By Liouville's theorem, the velocity distribution at the observer is the interstellar distribution evaluated on the same orbit, reduced by the survival probability:

$$f(\mathbf{r}, \mathbf{v}) = f_{\mathrm{LISM}}(\mathbf{v}_\infty)\, e^{-\beta_0 X}. \tag{3}$$

Here $\mathbf{v}_\infty$ is the velocity the atom had far from the Sun, which follows analytically from the orbit. This is the classical "hot model" of interstellar gas (Thomas, 1978; Lee et al., 2012). The beam flux seen by an observer is the integral of f over all arrival directions and speeds, done separately for direct and indirect atoms. It is computed numerically on a grid of directions and speeds, with the observer moving with Earth at $29.78\,\mathrm{km\,s}^{-1}$; the results change by less than 5×10^{-4} when the grid is refined.

2.2 Why the beam ratio measures the ionization rate

Each beam flux is proportional to the interstellar density n_{ISN} and, once detected, to the absolute efficiency ε. Its dependence on the ionization rate is described by the beam's mean exposure $\langle X \rangle = -\mathrm{d}\ln F/\mathrm{d}\beta_0$. For the ratio $R = F_{\mathrm{ind}}/F_{\mathrm{dir}}$ of the two beams this gives, to a good approximation,

$$\ln R = \mathrm{const} - \beta_0 \big(\langle X \rangle_{\mathrm{ind}} - \langle X \rangle_{\mathrm{dir}}\big). \tag{4}$$

The constant does not depend on n_{ISN} or ε, which cancel in the ratio, but β_0 remains. A higher ionization rate lowers the ratio, while a higher density leaves it unchanged.

2.3 Observing setup and hypotheses

IBEX and IMAP are Sun-pointing spinning spacecraft. IBEX-Lo looks perpendicular to the spin axis, so it scans a circle 90° from the Sun, which sees the

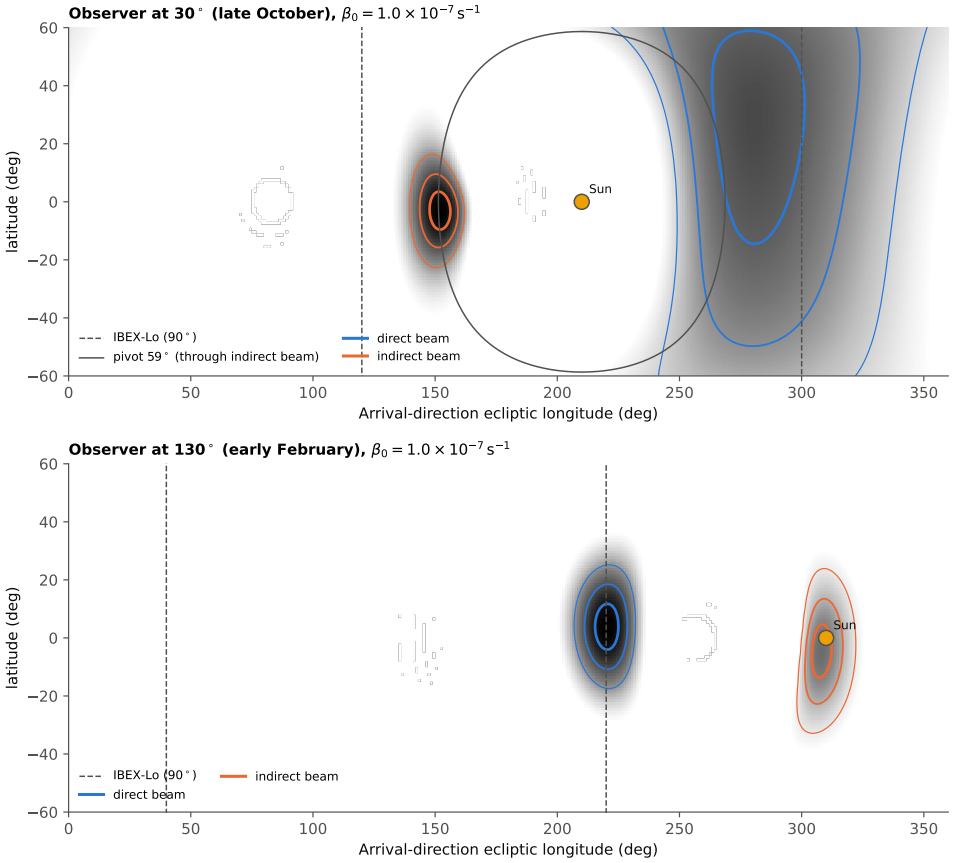


Figure 3: Helium flux on the sky seen from Earth in late October (top) and early February (bottom), in ecliptic coordinates of the arrival direction. Grey shading is the total flux; blue and orange contours trace the direct and indirect beams. The dashed curve is the IBEX-Lo viewing circle (90° from the Sun); the solid curve is the IMAP-Lo pivot angle that crosses the indirect beam.

direct beam well around February. IMAP-Lo can change this angle with its pivot platform (McComas et al., 2018; Kubiak et al., 2023). In the model, the angle needed to observe a beam is the angle between the beam centre and the Sun.

The direct beam is taken at its February peak (Earth at ecliptic longitude 130°), where IBEX-Lo has measured it for over a decade. The indirect beam is taken in late October (Earth at 30°), where it is bright and well separated from the direct beam. Two explanations of the SWAP result are compared:

- **H1**: the ionization rate is 1.43 times the model value, at the same density;
- **H2**: the density is 1.43 times higher, at the model ionization rate.

Both give the same pickup-ion production, so pickup ions alone cannot tell them apart.

3 Results

Two orbits to Earth. Figure 1 shows the two orbits that reach Earth in the two reference seasons, for a cold flow. The indirect orbits sweep 226° (October) and 234° (February) around the Sun, compared with 134° and 126° for the direct ones, and pass much closer to the Sun. That longer path near the Sun is the source of their larger ionization exposure.

Model check. The helium density along Earth's orbit (Figure 2) checks that the model behaves sensibly. Upwind, the model gives 0.011 cm^{-3} at the model ionization rate, close to the $\approx$0.01 cm^{-3} adopted by Swaczyna et al. (2025) for solar minimum. Downwind, solar gravity focuses the atoms into a cone where the density rises more than five-fold, and about 40% of the atoms there are on indirect orbits.

Where the beams appear. Figure 3 shows where the two beams arrive from. In late October the indirect beam is a compact spot about 59° from the Sun. The 90° IBEX-Lo circle misses it, but an IMAP-Lo pivot angle of 59° passes through its centre. In February the direct beam sits on the IBEX-Lo circle, while the indirect beam arrives from within a few degrees of the Sun and cannot be observed.

The beams over the year. Figure 4 follows both beams around Earth's orbit. The direct beam is bright all year. The indirect beam is comparable to it in November and December, fades through spring,

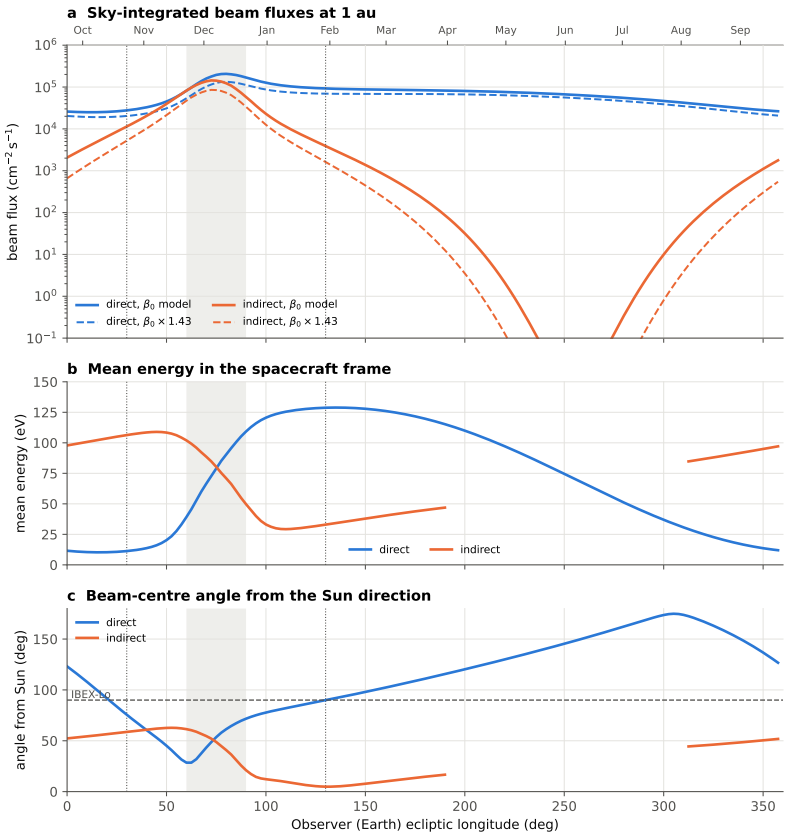


Figure 4: The two beams over the year, against Earth's ecliptic longitude (months on top). (a) Beam fluxes for the model ionization rate (solid) and a 43% higher rate (dashed). (b) Mean beam energy seen by the detector. (c) Angle between the beam centre and the Sun. Grey bands mark seasons where the two beams overlap on the sky; dotted lines mark the two reference seasons.

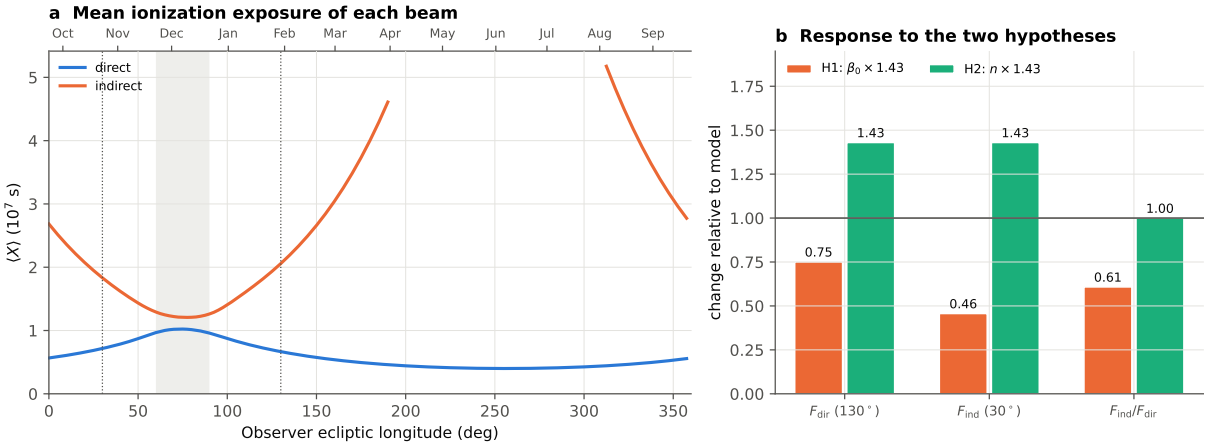


Figure 5: (a) Mean ionization exposure $\langle X \rangle$ of each beam over the year. (b) Change of the February direct-beam flux, the late-October indirect-beam flux and their ratio under the two hypotheses, relative to the model.

and is negligible around June. From late November to late December the two beams overlap on the sky and cannot be separated. In autumn the indirect beam arrives at 100 eV to 110 eV, compared with ≈130 eV for the February direct beam. It stays 50°–60° from the Sun, so it needs a pivot angle well below 90°. In every season the indirect beam responds more strongly to the ionization rate (panel a).

The ratio separates the hypotheses. This is the main result (Figure 5). The indirect beam accumulates 1.5 to 5 times more exposure than the direct beam, with the largest contrast early in autumn. For the reference pair the mean exposures are 0.67×10^7 s (direct, February) and 1.83×10^7 s (indirect, October). Under H1, the higher ionization rate lowers the direct beam to 0.75 and the indirect beam to 0.46 of the model, so the ratio drops to 0.61. Under H2, both

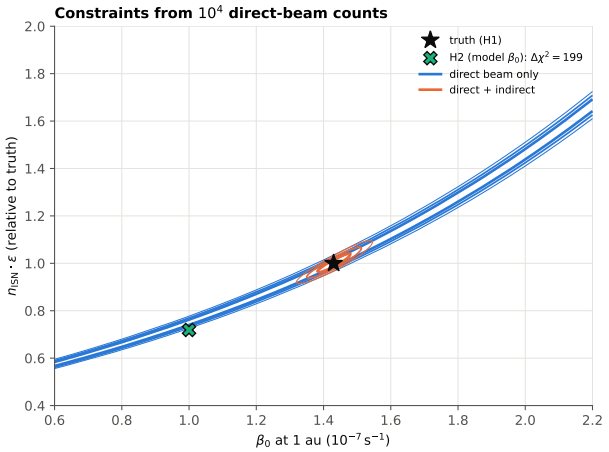


Figure 6: Constraints on the ionization rate β_0 and on the product of density and detector efficiency, $n_{\mathrm{ISN}}\varepsilon$, from noise-free simulated counts generated under H1 (star). Blue: direct beam only; orange: direct and indirect beams together. Contours are the 1, 2 and 3σ levels. The cross marks H2.

beams rise by 1.43 and the ratio stays exactly the same. The absolute direct-beam flux alone could only separate the two if the detector efficiency were known to much better than the 25–43% difference between them. The ratio does not need it.

Breaking the degeneracy. Figure 6 shows the same point as a fit. Noise-free counts are simulated under H1, with 10^4 counts in the February direct beam and about 750 in the October indirect beam for the same exposure time. They are then fitted for β_0 and the unknown product $n_{\mathrm{ISN}}\varepsilon$, using a Poisson likelihood (Baker & Cousins, 1984). The direct beam alone only constrains a curved band: a higher ionization rate can always be traded for more helium or a more efficient detector. Adding the indirect beam closes the contour around the true values, and H2 is strongly disfavoured ($\Delta\chi^2 \approx 199$). This idealized test shows how the second beam lifts the degeneracy. It is not a forecast of the achievable precision, which depends on the effects discussed next.

4 Limitations and future work

The model is a feasibility study of the size of the beam-ratio signal. A full analysis of IMAP-Lo data would need the following, roughly in order of importance.

Radiation pressure. The orbits are purely Keplerian. Kowalska-Leszczyńska et al. (2023) showed that radiation pressure changes the IMAP-Lo helium fluxes by up to 9% for the indirect beam, likely more than the statistical uncertainty. Including it requires numerical orbits, since the closed form of Equation (2) no longer holds.

Other near-Sun losses. The ratio measures the *total* extra loss of indirect atoms, not photoionization alone. Elastic scattering by solar-wind particles has been proposed as an alternative to a higher ionization rate (arXiv:2409.11784, 2024), and it would lower the ratio in the same way. Separating the two would rely on their different energy and angular signatures.

Secondary helium. The Warm Breeze (Kubiak et al., 2014) arrives from a different direction and has its own direct and indirect beams, which must be added to the model.

Time-dependent and non-r^{-2} ionization. Indirect atoms spend months near the Sun, and the two beams are observed months apart, so solar-cycle changes of the ionization rate matter. Electron-impact ionization does not fall off as r^{-2} (Sokół et al., 2019). Both effects need a numerical exposure integral along each orbit.

Instrument response. The model uses sky-integrated fluxes. A realistic prediction needs the IMAP-Lo collimator, the pivot schedule (Kubiak et al., 2023) and the energy-dependent detection efficiency. Since the two reference beams arrive about 20% apart in energy, an energy-dependent efficiency does not fully cancel in their ratio. Choosing pairs of observations at matched energies, as recommended by Bzowski et al. (2023), would reduce this effect. A first look at a simple energy threshold is in progress.

Inflow-parameter uncertainties and counting statistics. The uncertainties of the inflow speed, direction and temperature (McComas et al., 2015) also move the beam ratio, and the number of counts needed to detect the signal depends on the observing season. These are being estimated. Since the inflow parameters are derived from direct-beam data and are correlated, the natural approach is a joint fit of the inflow parameters and the ionization rate to both beams.

5 Summary

A simple analytic model of interstellar helium at 1 au confirms that the ratio of the indirect to the direct beam is a sensitive, calibration-free measure of the helium ionization rate. If the ionization rate is 43% higher than models predict, as New Horizons/SWAP pickup ions suggest, the ratio of the late-October indirect beam to the February direct beam drops by about 40%. A 43% higher interstellar density leaves the ratio unchanged. The indirect beam is bright and well separated in autumn, about 60° from the Sun, where IMAP-Lo's pivot platform can observe it. Radiation pressure, additional near-Sun losses, secondary helium and the instrument response are the main effects to add before the method can be applied to real data.

Code availability

The model, the scripts that generate the data and the notebook that makes the figures are in the project repository (`src/isnhe`, `scripts/`, `notebooks/plot_figures.ipynb`).

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